

Spatio-temporal data visualization

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How to visualize spatio-temporal (ST) data in order to depict dynamics of a phenomena ?

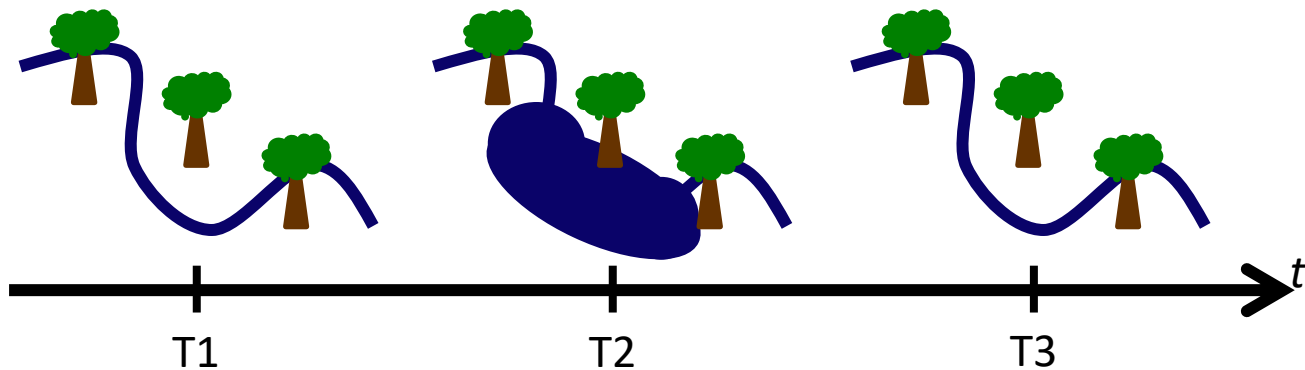
Representing change in space

- Representing the result of the change
Evolution of spatial entities along time
- Representing the phenomena at the origin of the change
Representing events, process, limited in time and space

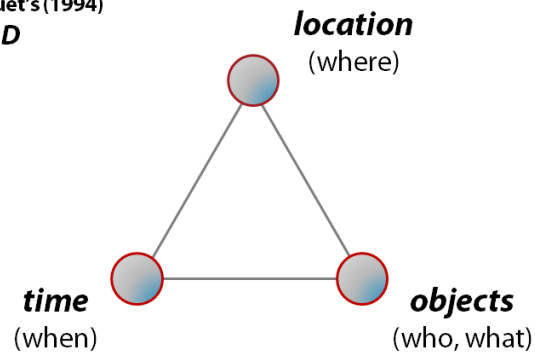
Representing change in space

- Representing the result of the change
Evolution of spatial entities along time

Temporal evolution of a river



Peuquet's (1994)
TRIAD

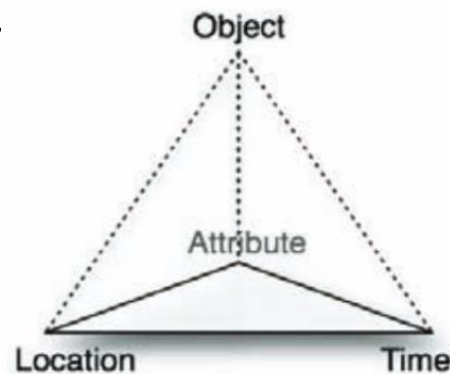
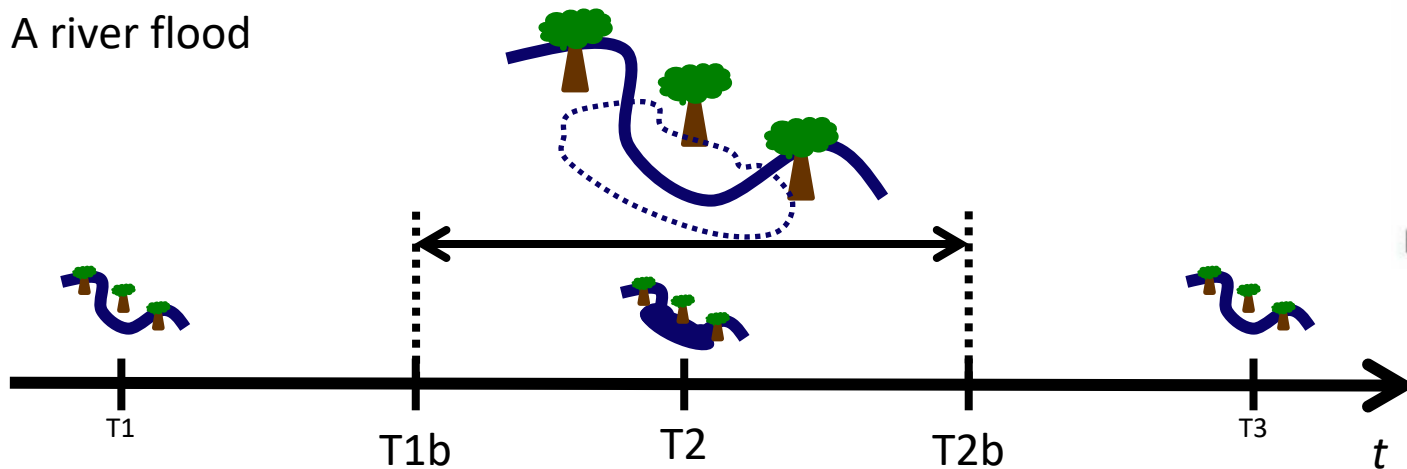


Representing change in space

- Representing the phenomena at the origin of the change

Representing events, process, limited in t

A river flood



Pyramid model
Mennis et al., 2000

Spatio-temporal data

- **Representing the result of the change**

Temporal series: each entry represents the temporal state of an atemporal spatial entity

- Car positions
- Temperature grids
- Demographic data at different dates

Spatio-temporal data

- **Representing the phenomena at the origin of the change**

Event series: each entry represents an event limited in time and space

- Car travel
- Fire departure

Analyzing ST data

- Change affecting objects along time
 - Change of value/state/nature/position/geometry
- Spatio-temporal dynamics which structure a phenomena
 - ST distribution of events
 - Duration, trajectories

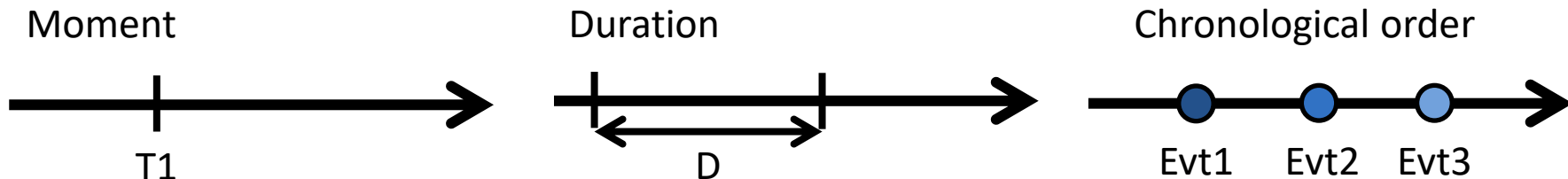
ST data analysis

- Event level
 - temporalities/position of specific event
- Phenomena level
 - Identifying pattern
 - Localized trends
 - Clusters
 - Propagation axes

ST data analysis

Temporalities

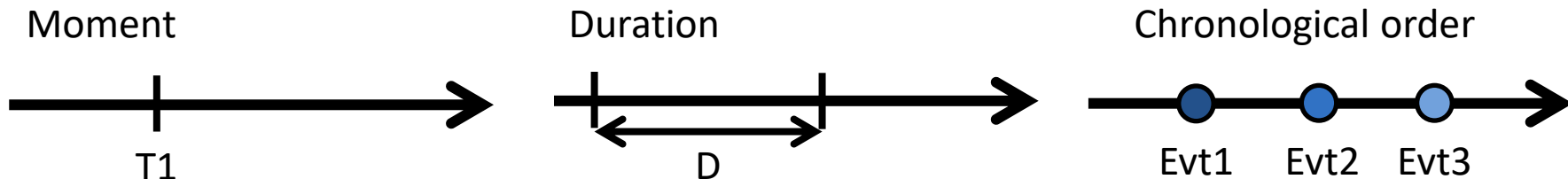
Hewitt, 1983, Allen, 1983, DiBiase et al., 1992,
MacEachren, 1994



ST data analysis

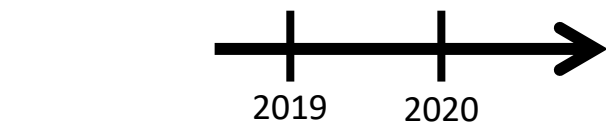
Temporalities

Hewitt, 1983, Allen, 1983, DiBiase et al., 1992,
MacEachren, 1994

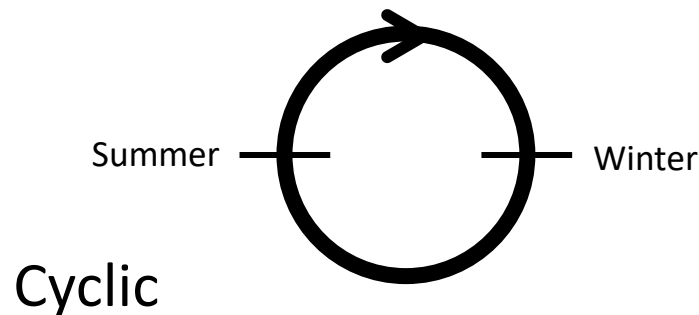


Temporal structure

Kraak, 2013

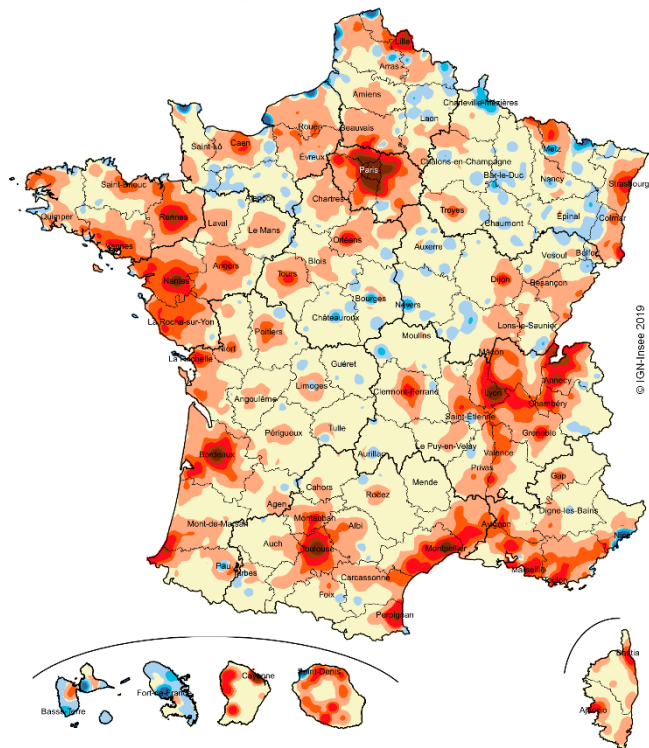


Linear

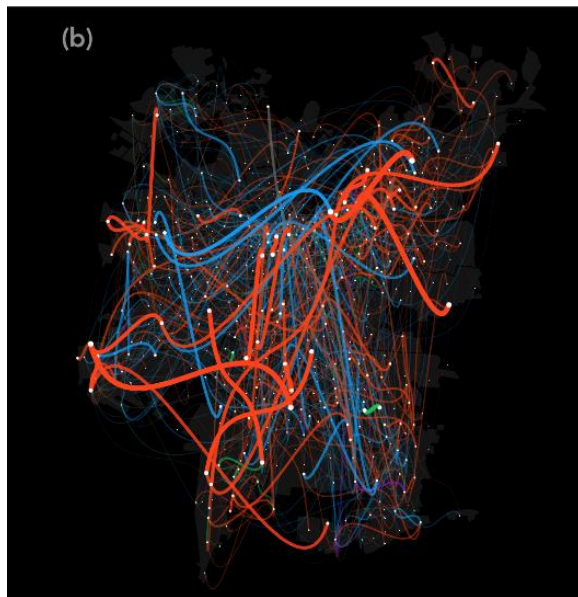


Cyclic

Implicit visualization of time



Source : Insee, recensements de la population de 2007 et 2017. <https://www.insee.fr/fr/statistiques/4267787>



Visualization...

- Of flows
- Of events appearing
- Of value changes

...for a given period

Pérez-Messina, I., Graells-Garrido, E., Lobo, M. J., & Hurter, C. (2020). Modalflow: cross-origin flow data visualization for urban mobility. *Algorithms*, 13(11), 298.

Direct encoding of time

Representation of temporalities through visual channels

Direct encoding of time

Representation of temporalities through visual channels

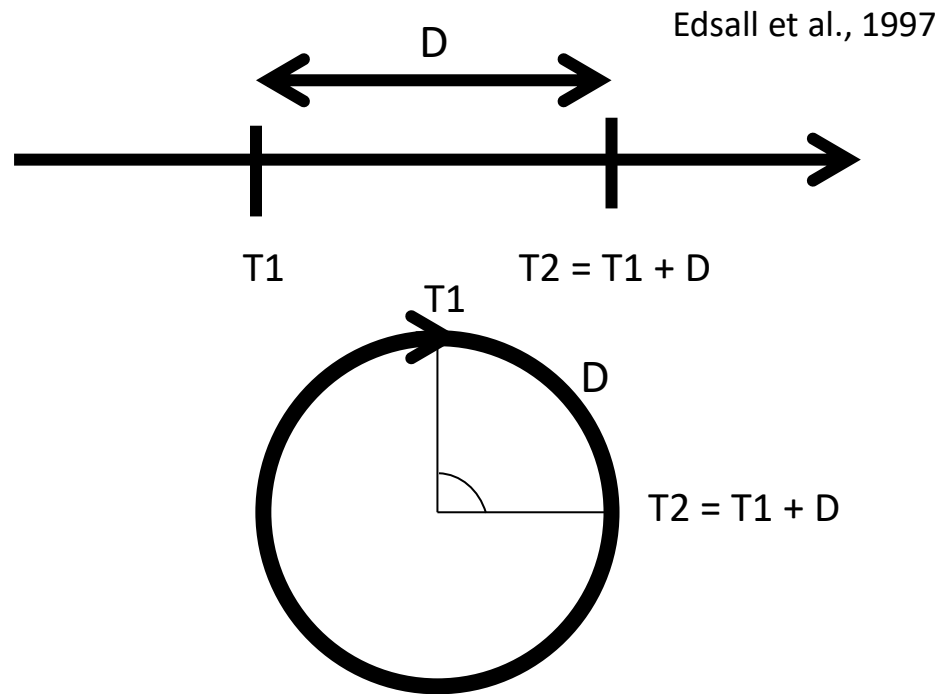
Graphic position

- timeline

linear time

- timewheel

cyclic time



Direct encoding of time

Covid 19 visualization dashboard

<https://mapthenews.maps.arcgis.com/apps/opsdashboard/index.html#/5df19abcf8714bc590a3b143e14a548c>

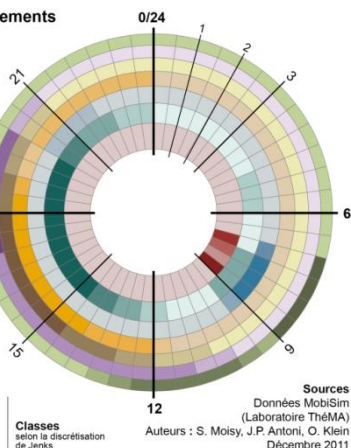
Nombre de déplacements par motif

Départs tous modes confondus par tranches de demi-heure

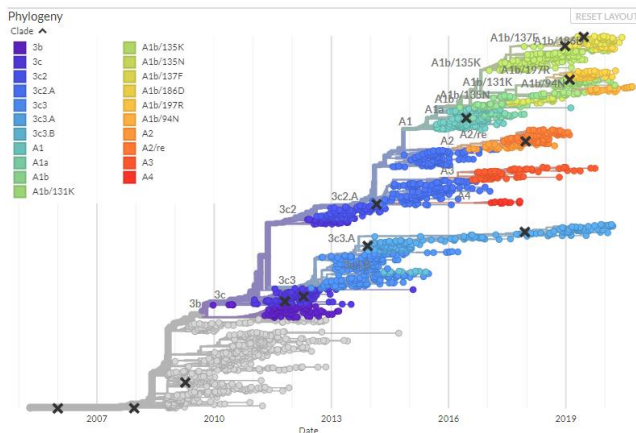
Les surfaces des unités temporelles de 30 min, sont égales sur chaque disque

Motifs de déplacement

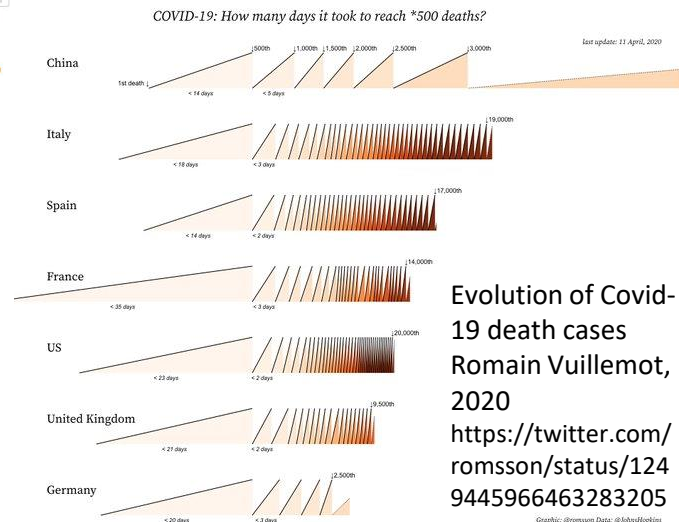
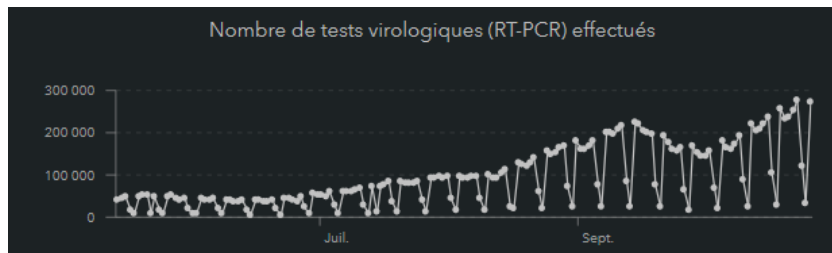
Etudes Réseau soc. Achats Loisirs Travail Retour Accomp.



Antoni et al. 2012



Nextstrain environment
<https://nextstrain.org/ncov/global>



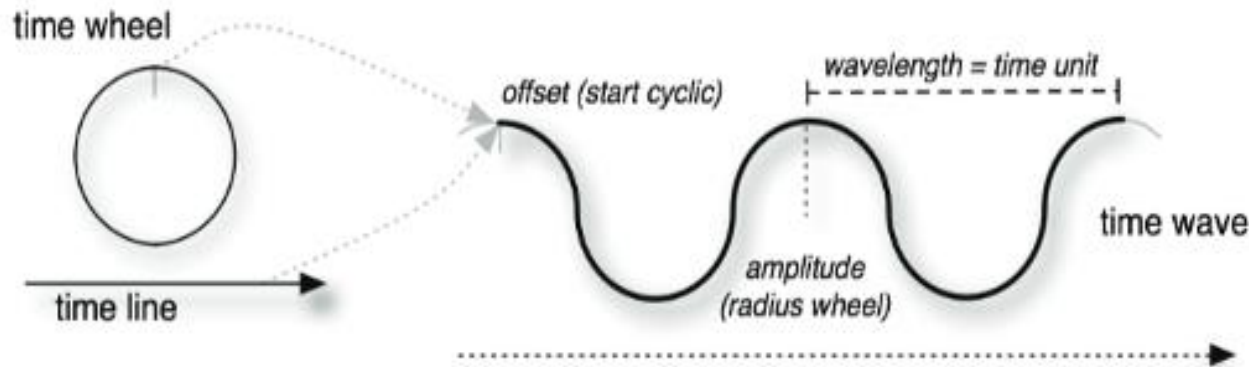
Evolution of Covid-19 death cases
Romain Vuillemot, 2020
<https://twitter.com/romsson/status/1249445966463283205>

Graphic: @romsson Data: @JohnsHopkins

Direct encoding of time

Mixt diagrams representing

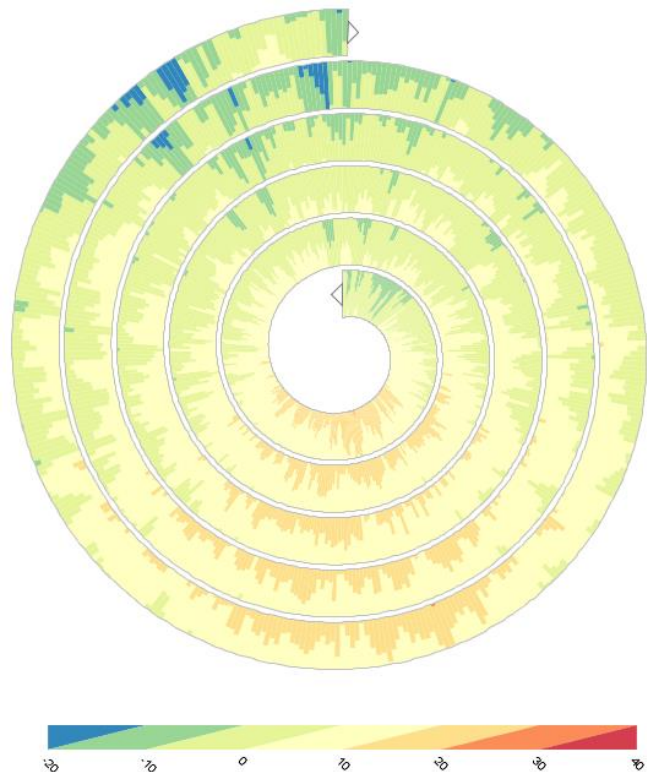
- Linear time
- Through a succession of cyclic instances



Time Wave
Li, 2010

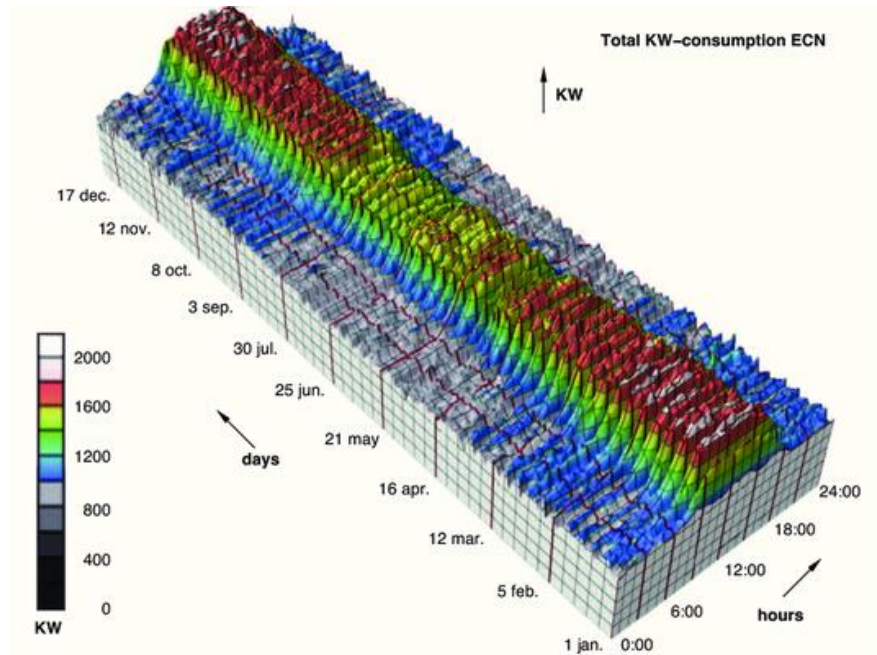
Direct encoding of time

Highlighting cyclic structures



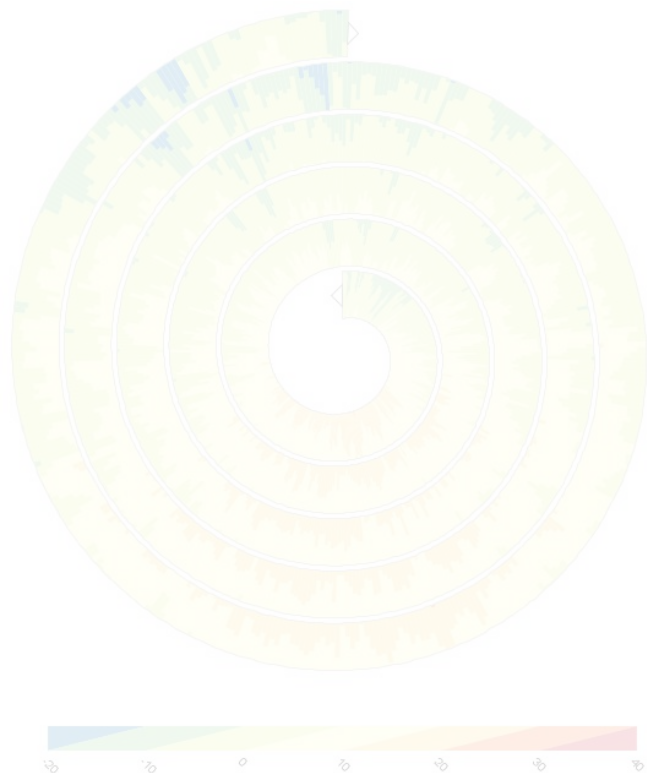
Enhanced Interactive
Spiral
Tominski et al., 2008

Cluster Calendar
Van Wijk, 1999



Direct encoding of time

Highlighting cyclic structures

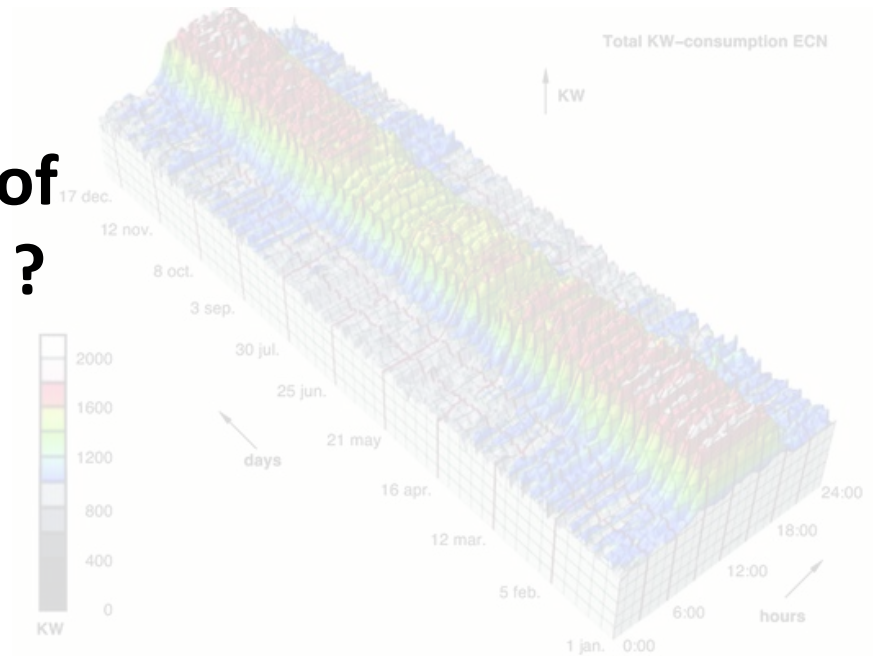


Enhanced Interactive
Spiral

Tominski et al., 2008

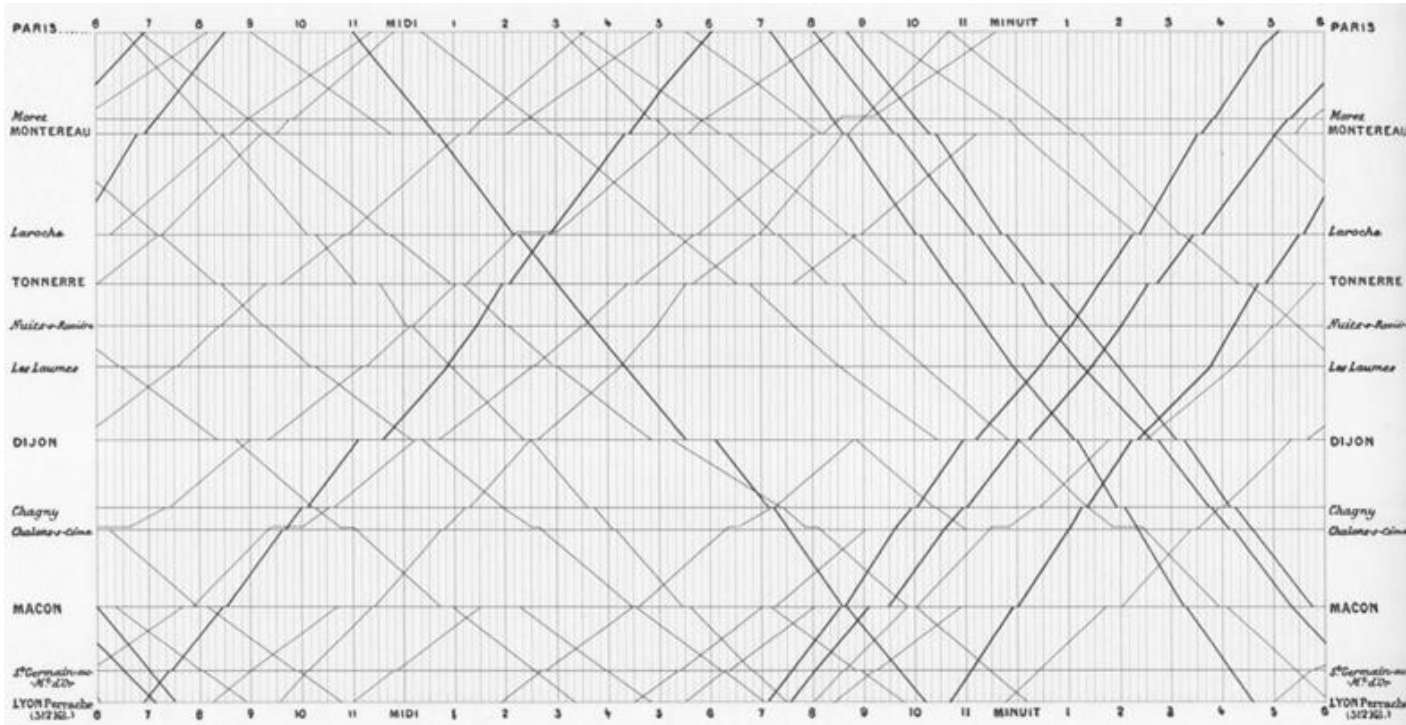
**What of
space ?**

Cluster Calendar
Van Wijk, 1999



Direct encoding of time

Do we need two dimensions ?

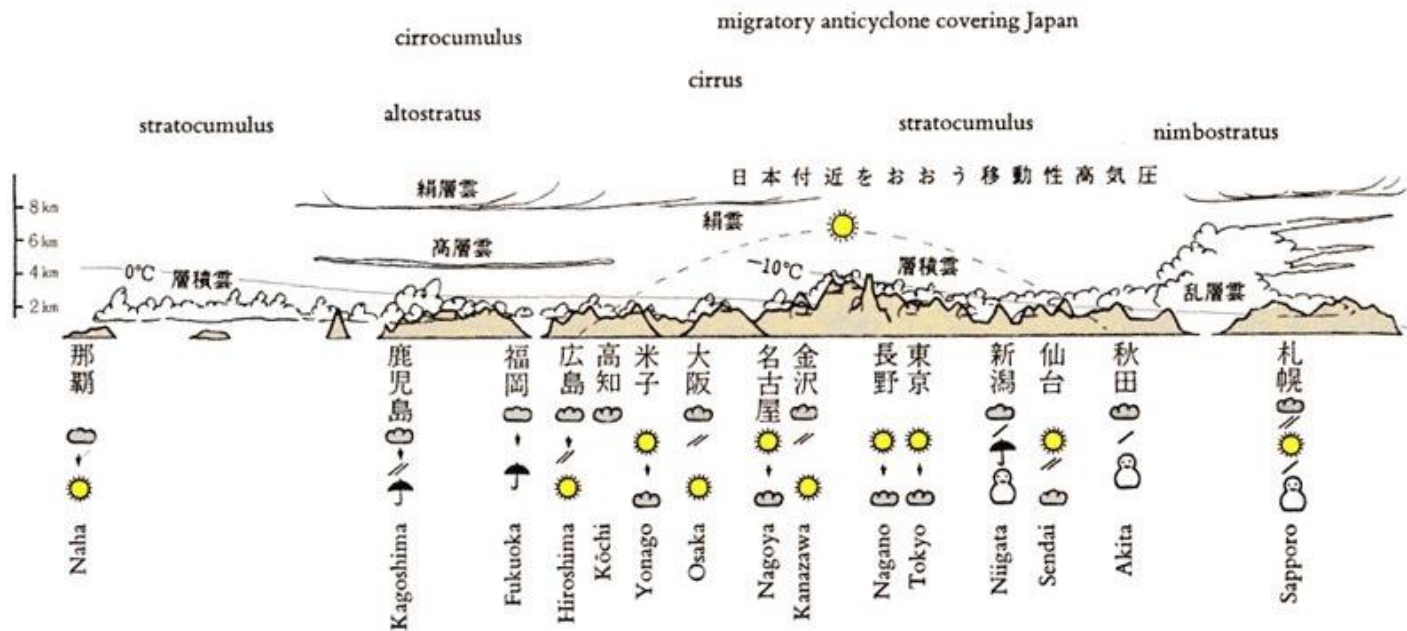


Train schedule of
Marey (1885)
Tufte (1990)

Direct encoding of time

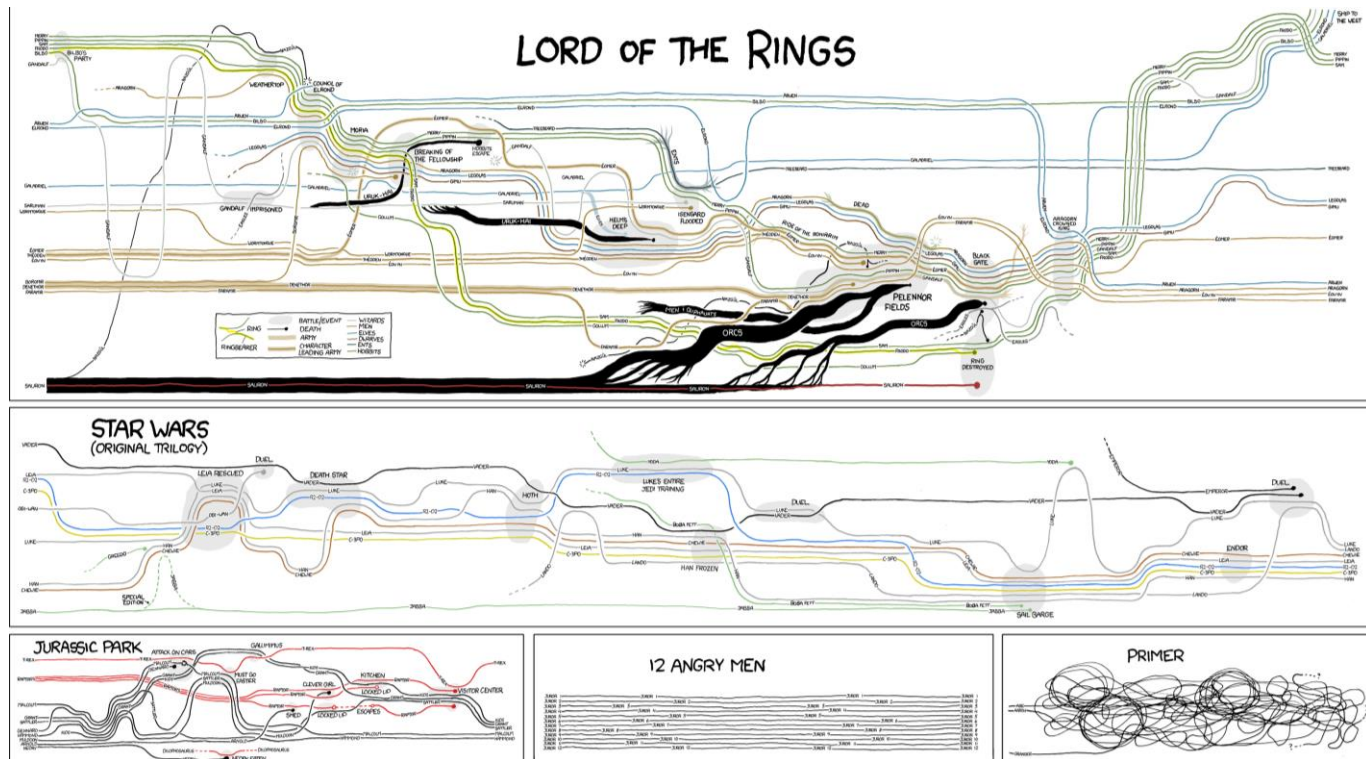
Do we need two dimensions ?

Redrawn from *Akahata* [Red Flag], Tokyo, March 7, 1985. English translation added here. Note the changing weather reported for some cities, for example, Sapporo at the far right of the map.



Direct encoding of time

Do we need two dimensions ?



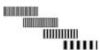
Narrative Charts

XKCD

<https://xkcd.com/657/>

Direct encoding of time

2D map

	Moment	Duration	Return period	Chronological order	Rate of change
Taille 		+++	++	++	+++
Valeur 		+++	++	++	+++
Grain 	+++	+++	++	++	+++
Couleur (teinte) 	+		(+)	(+)	
Orientati-on 	+++		++	++	
Forme 	+++		++	++	

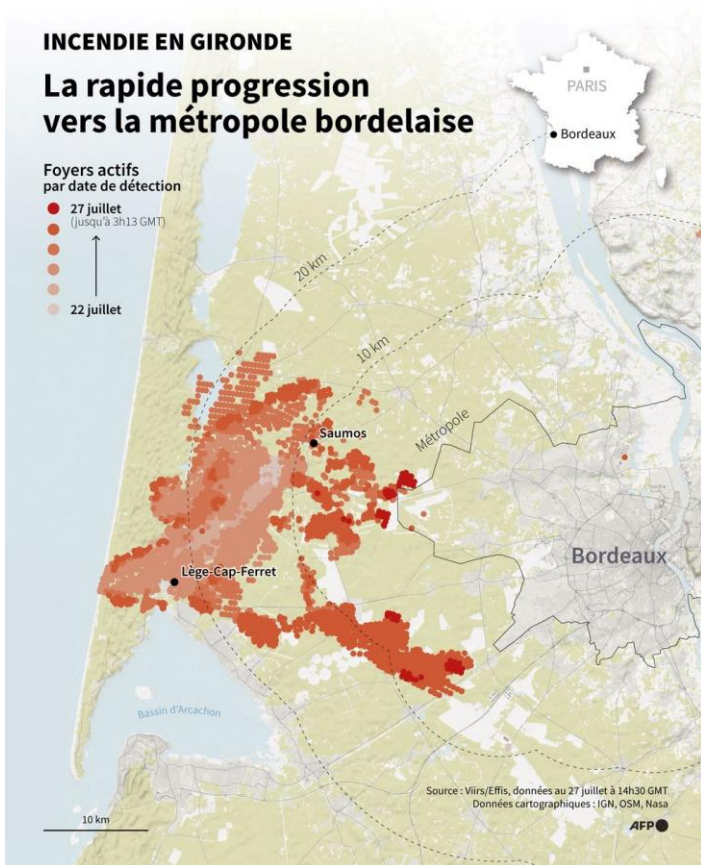
Direct encoding of time

INCENDIE EN GIRONDE

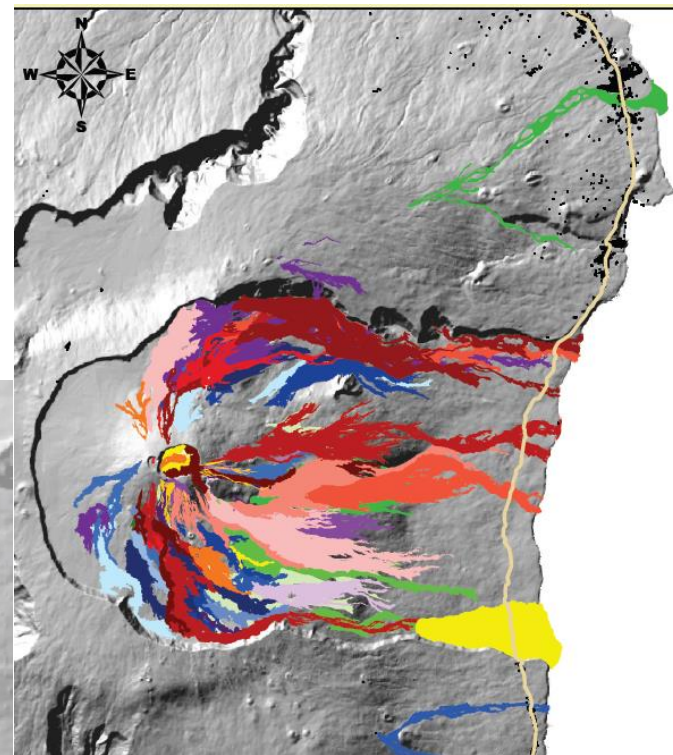
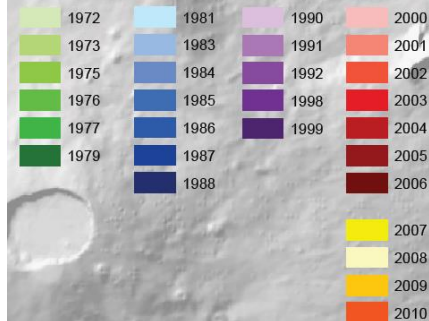
La rapide progression vers la métropole bordelaise

Foyers actifs
par date de détection

- 27 juillet
(jusqu'à 3h13 GMT)
- 22 juillet

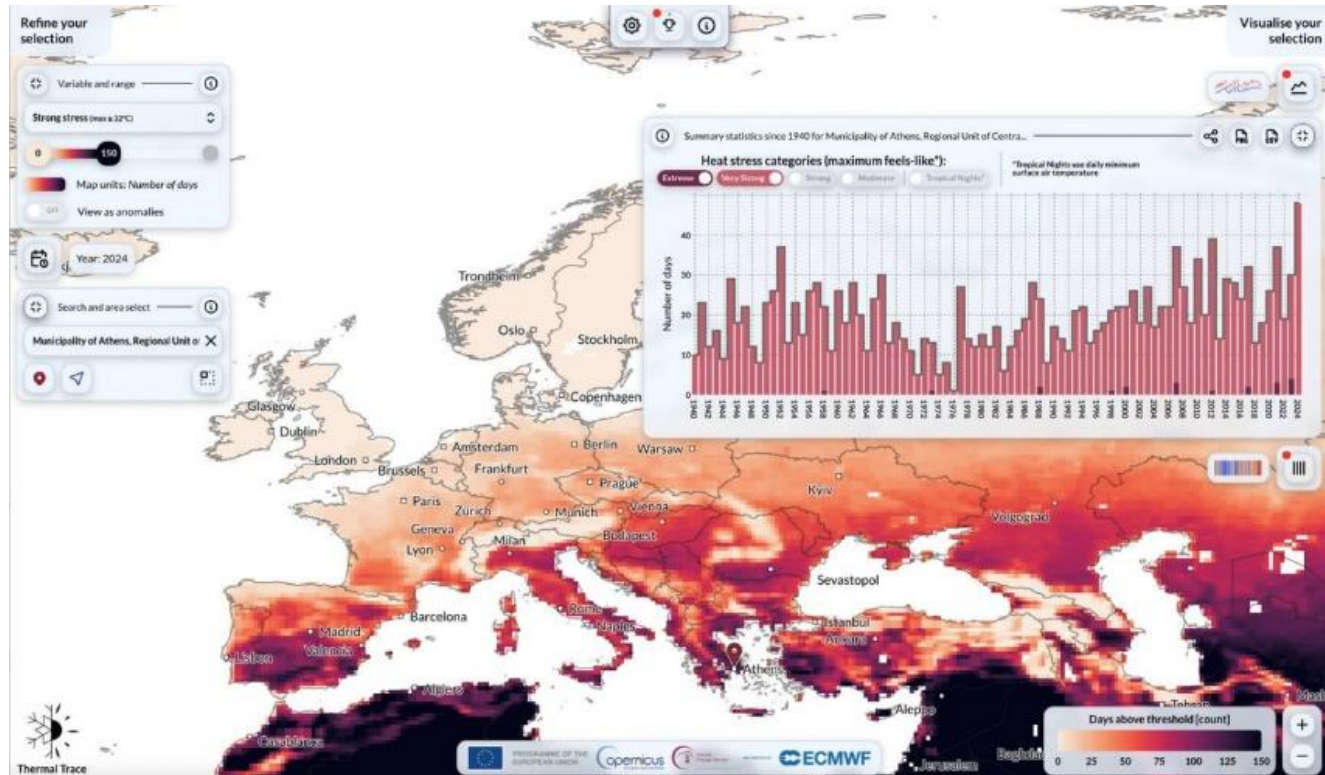


Coulées de lave par date (a, b)



Direct encoding of time

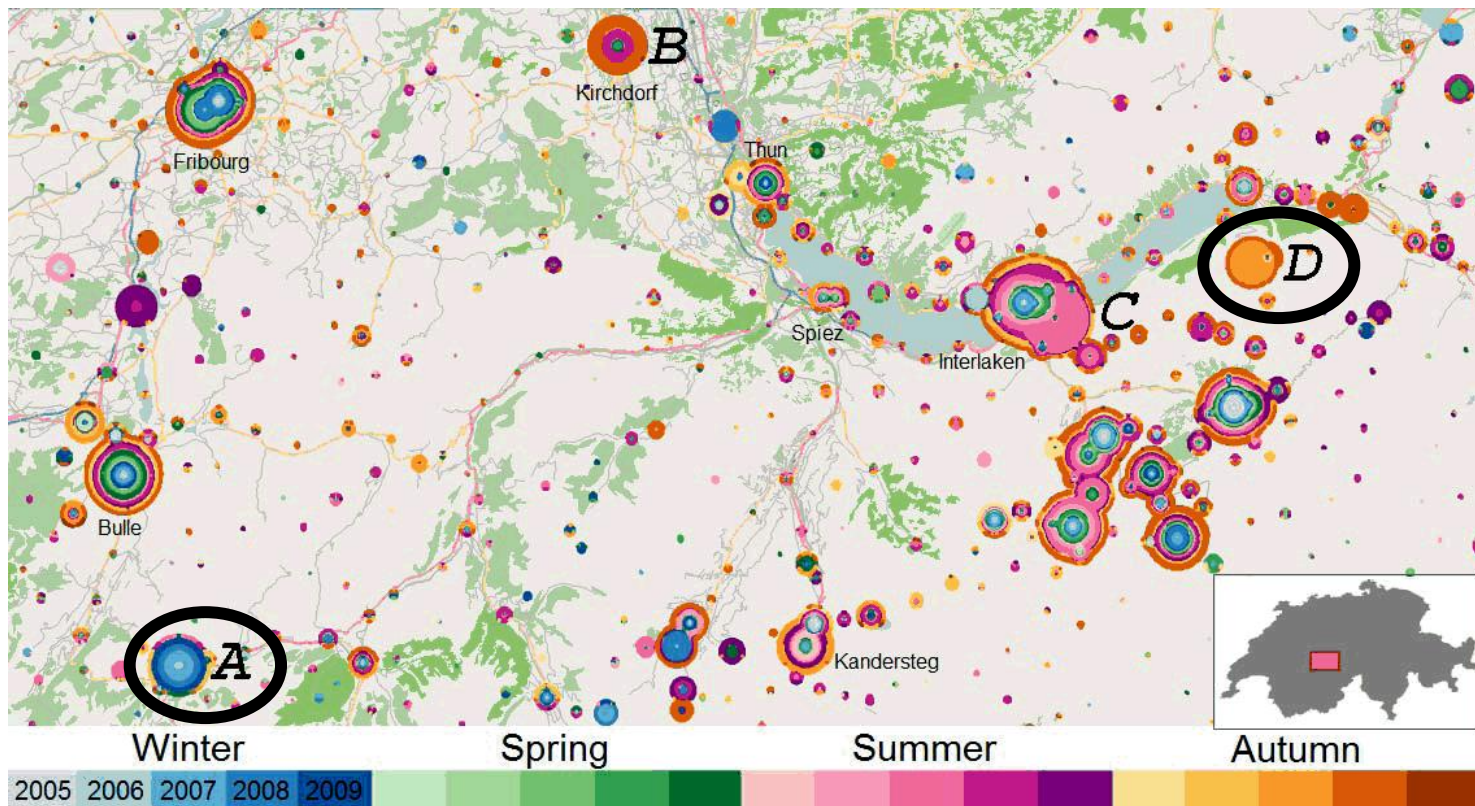
2D map



Direct encoding of time

D: Cluster
Autumn 2007

A: Cluster
Winter

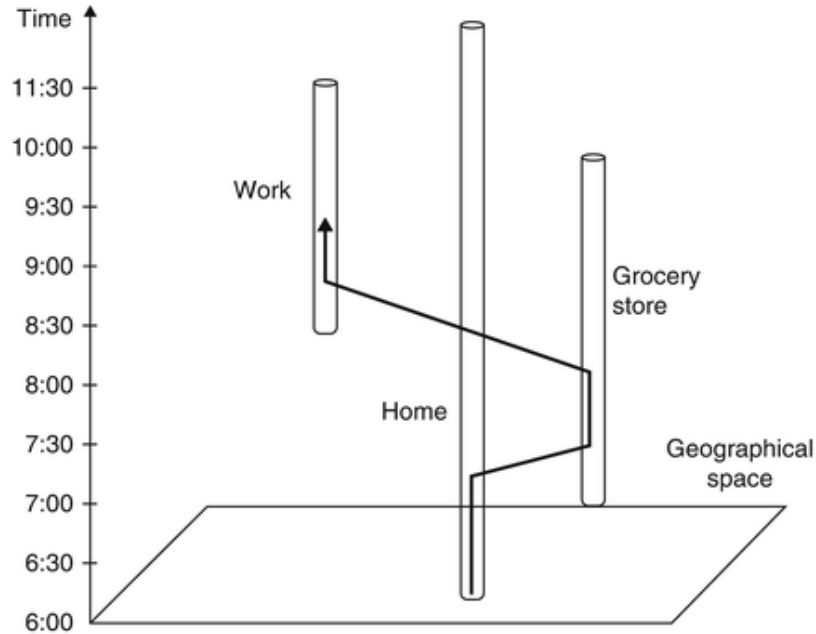


Growth Ring Maps

[Bak et al., 2009, Andrienko et al., 2011]

Direct encoding of time

Space-Time Cube

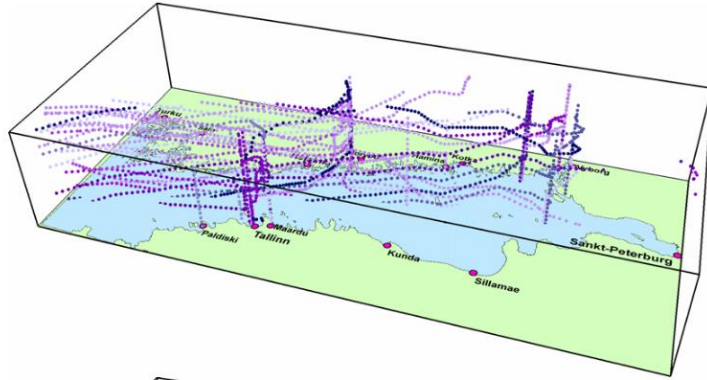


Hagerstrand, 1970, *Time Geography*

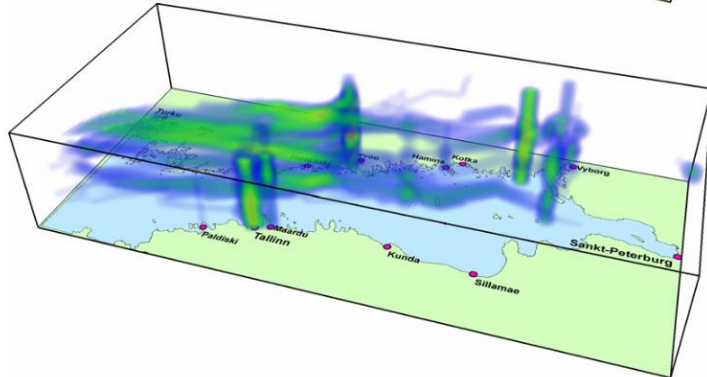
- X,Y
→ geographical space
- Z
→ time

Encodage direct du temps

Space-Time Cube



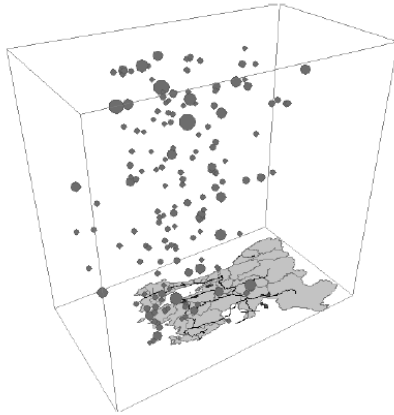
ST data distribution
– trajectories



Andrienko, G., Andrienko, N., Demsar, U., Dransch, D., Dykes, J., Fabrikant, S. I., ... & Tominski, C. (2010). Space, time and visual analytics. *International journal of geographical information science*, 24(10), 1577-1600.

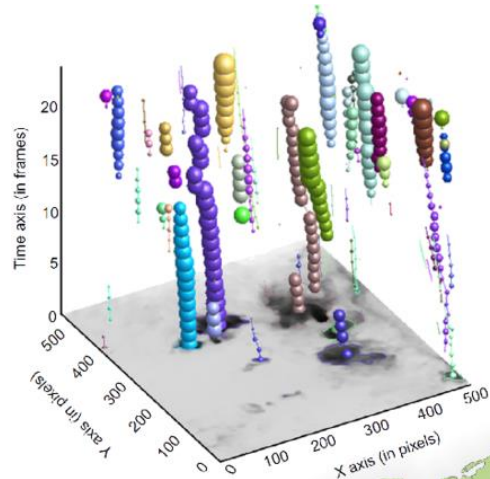
Encodage direct du temps

Space-Time Cube: Projection du temps sur un 3eme axe



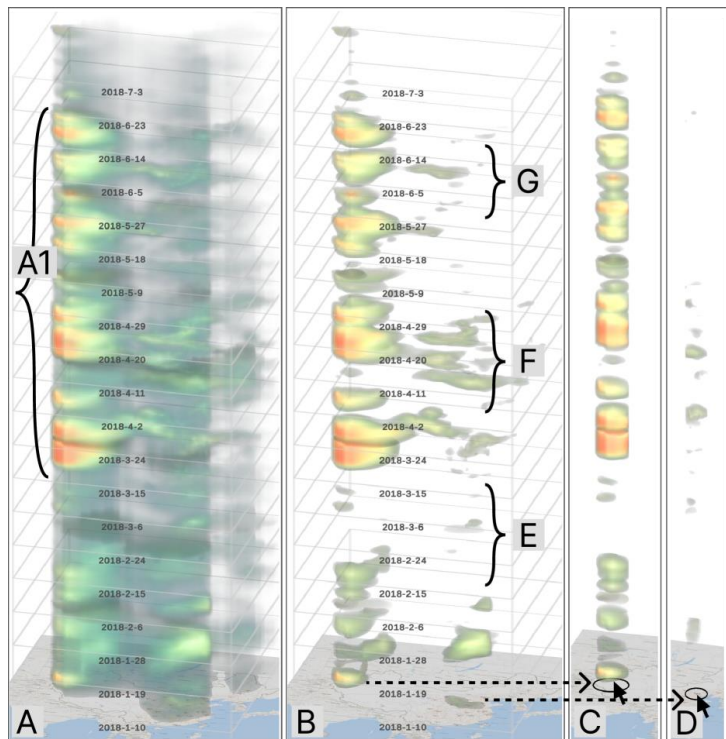
ST data distribution

- trajectories
- Punctual events



Encodage direct du temps

Space-Time Cube: Projection du temps sur un 3eme axe

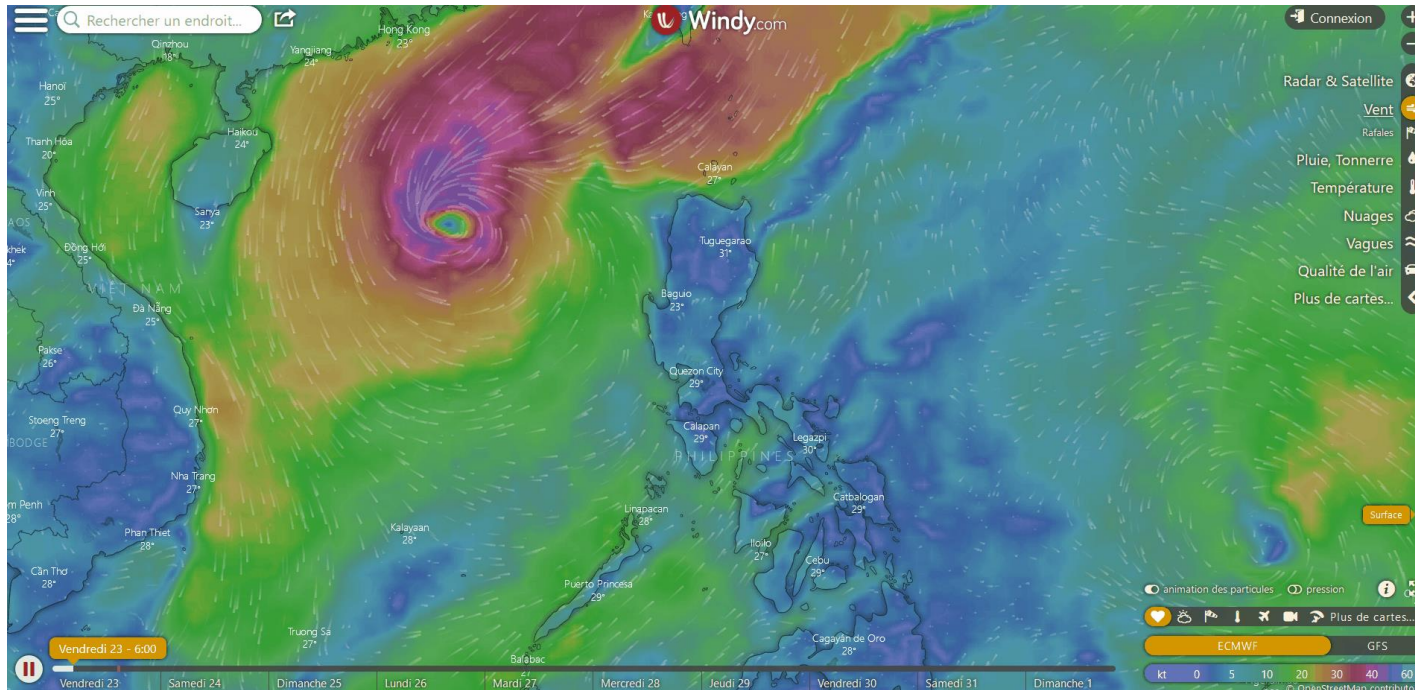


ST data distribution

- trajectories
- Punctual events
- 2D+T grids

Temporal animation

<https://www.windy.com/>



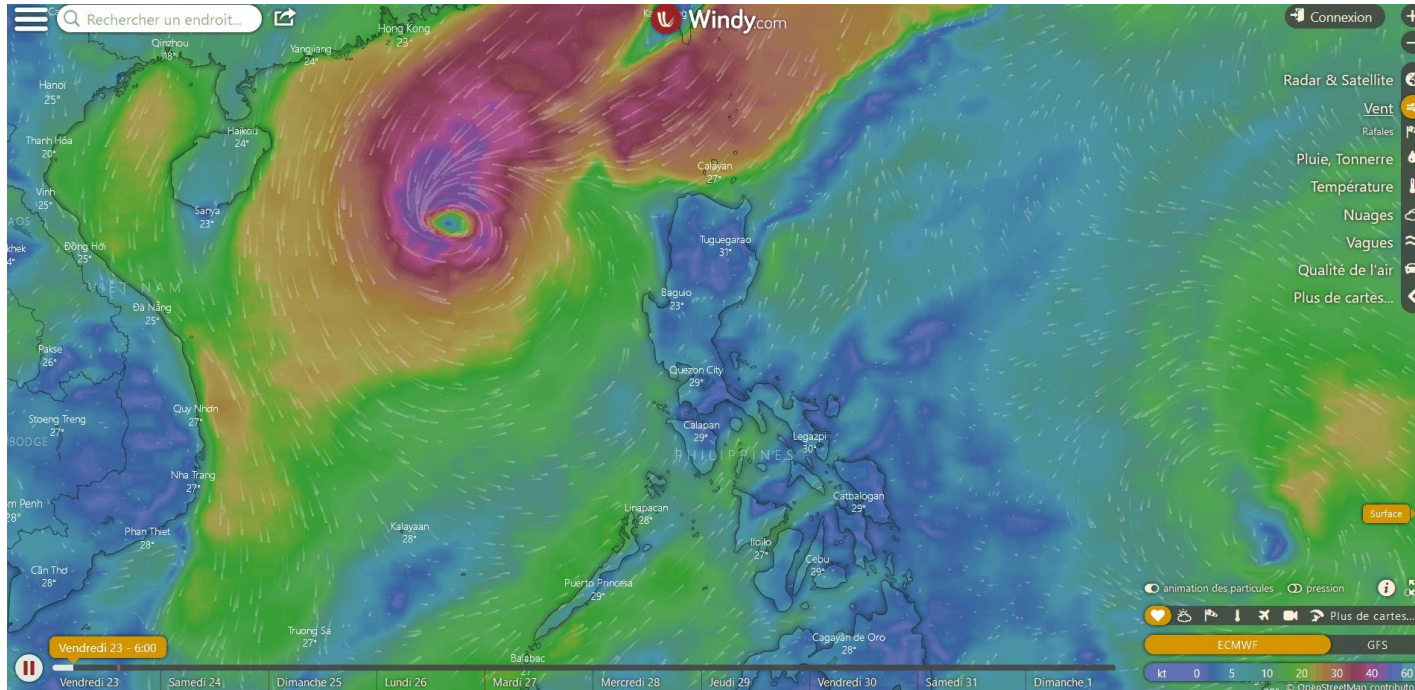
Two types of animation here:

- **Animated glyph** encoding wind direction and speed (**dynamic visual variable**).
- **Temporal animation**. Different time frames represent different moments in the dataset

Allows to visualize different temporal sequences

Temporal animation

<https://www.windy.com/>



Efficient to show spatio-temporal dynamics

Impactful

Two types of animation here:

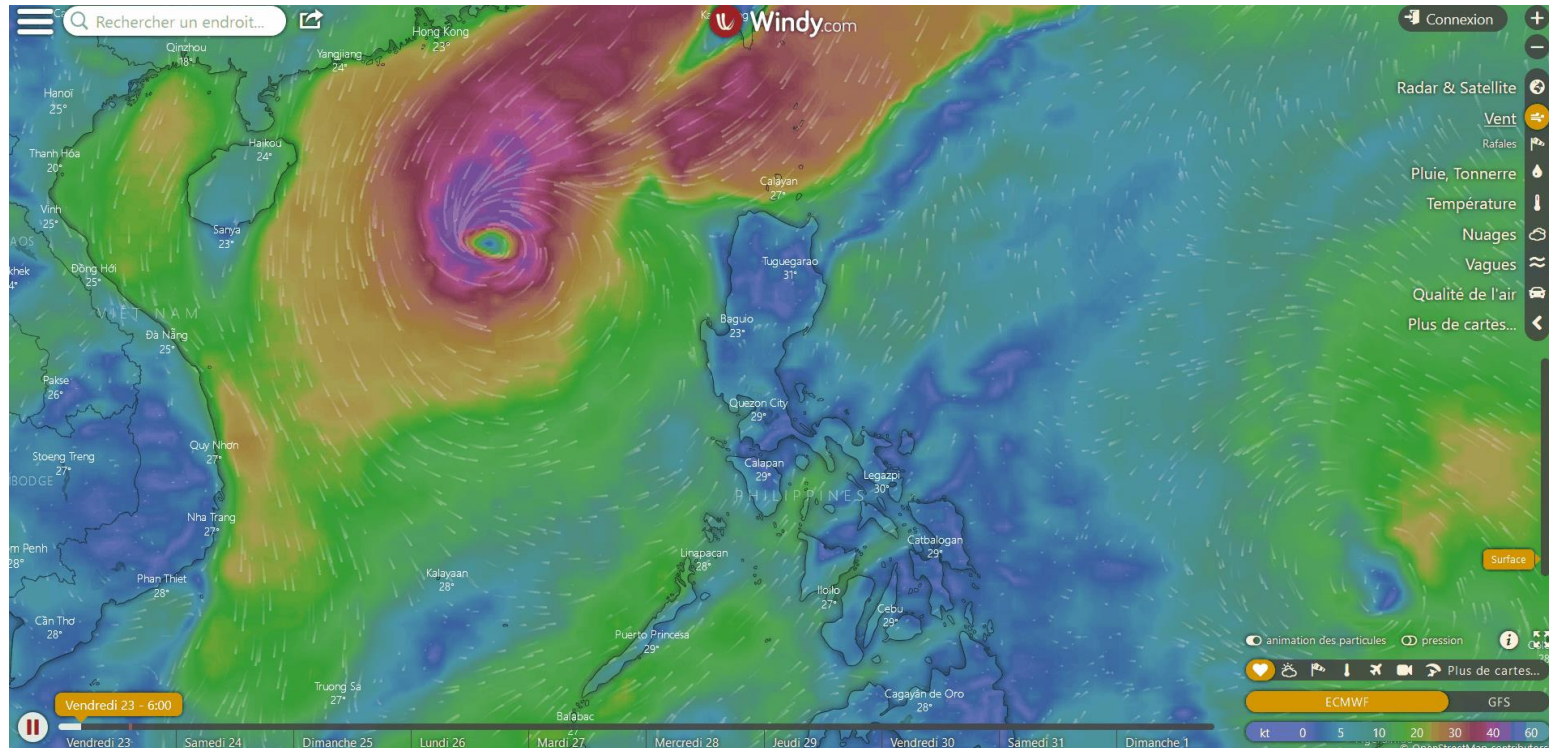
- **Animated glyph** encoding wind direction and speed (**dynamic visual variable**).
- **Temporal animation**. Different time frames represent different moments in the dataset

Allows to visualize different temporal sequences

Temporal animation

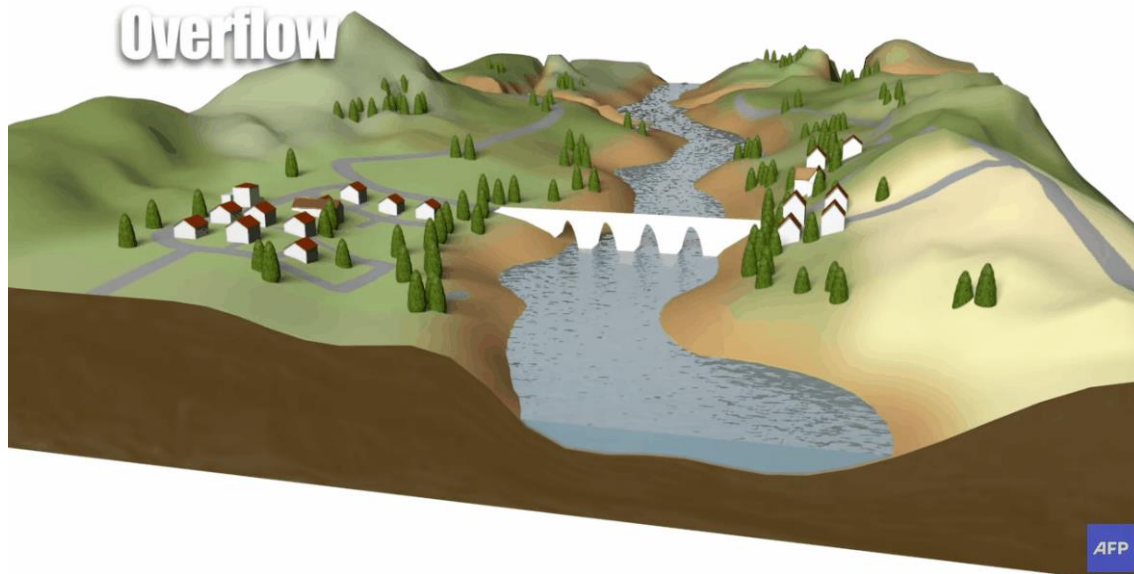
<https://www.windy.com/>

Animation time → linear representation of time



Animation time

Animation time → unlinear succession of chronological phases



The causes of
flooding
AFP
<https://www.youtube.com/watch?v=GQ-iWUkg4t0>

Temporal animation

Effective for interpreting the spatiotemporal dynamics of a phenomenon (movement, rate of change) and showing micro-stages. Impactful for the general public.

But...

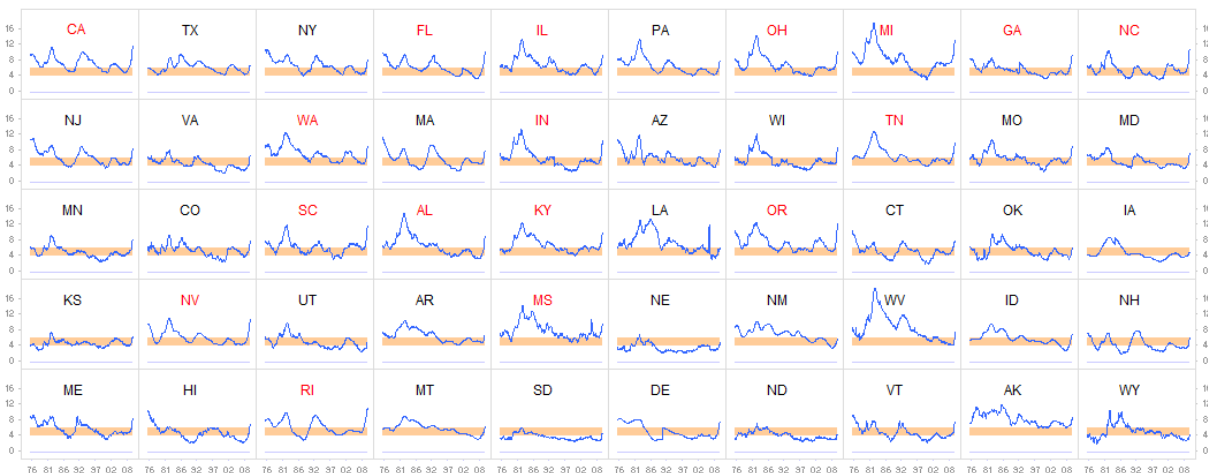
- Not the most efficient for comparing two temporal states
→ It is easier to compare two views side by side than to have to remember one of the two.
- Not efficient for a global analysis of a ST phenomena
→ The user must memorize the various sequences. Working memory is limited; when these limits are reached, the cognitive load becomes excessive, especially if many things are changing simultaneously.

Importance of interaction to control the animation

Small multiples: juxtaposition

Juxtaposing time

Monthly Unemployment Rates by State, Jan 1976 - Apr 2009



Source: Bureau of Labor Statistics

Notes: The orange band denotes a "normal" unemployment rate (4%-6%);
State code in red: unemployment rate in April 2009 is higher than the US average

Collection of temporal
representations

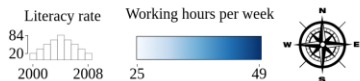
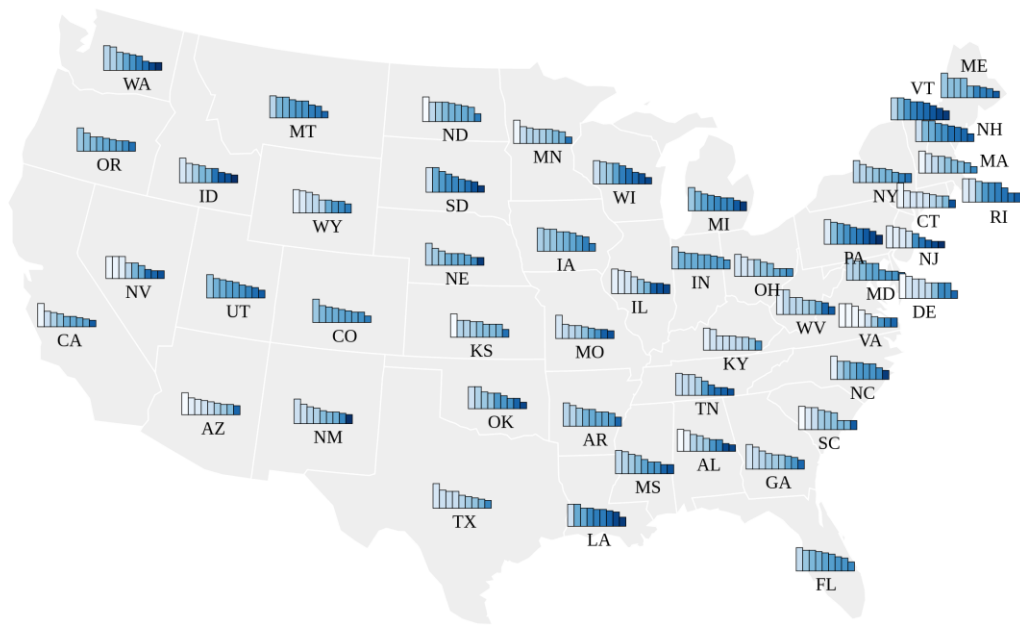
→ one per spatial location

Better Know a Visualization: Small Multiples Zach Gemignani, 2010

<https://www.juiceanalytics.com/writing/better-know-visualization-small-multiples>

Small multiples: juxtaposition

Juxtaposing time

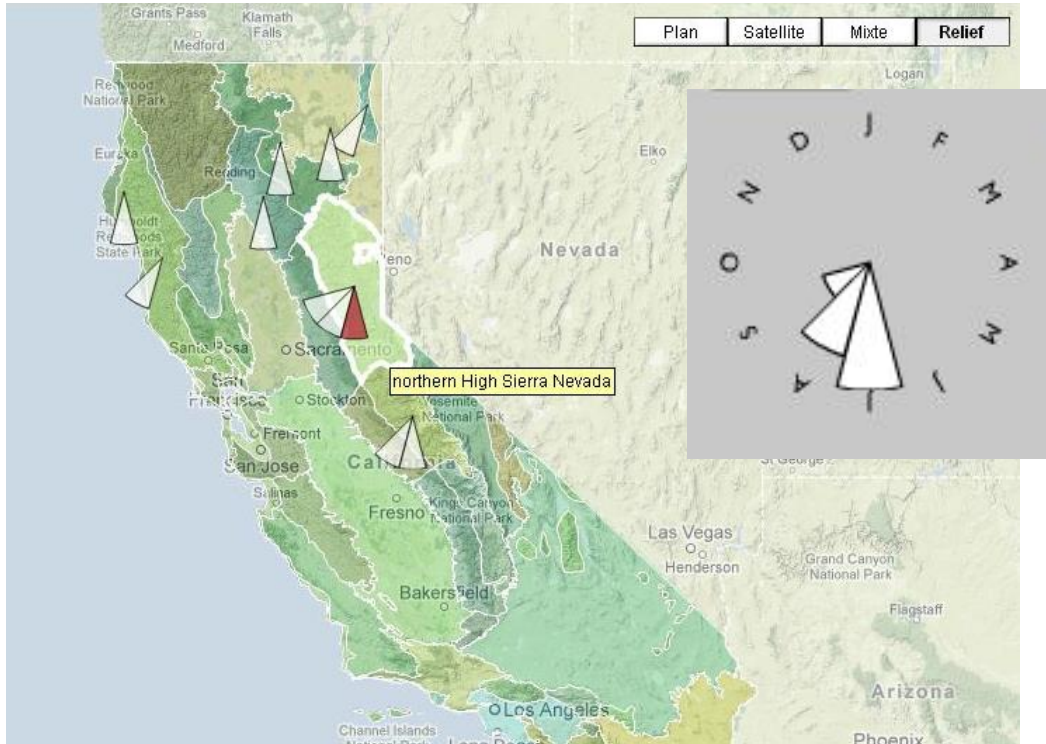


Integrating temporal representations into map through glyphs

Peña-Araya, Vanessa, Emmanuel Pietriga et Anastasia Bezerianos. "Une comparaison des visualisations pour identifier la corrélation dans l'espace et le temps." IEEE Transactions on Visualization and Computer Graphics 26.1 (2019): 375-385.

Small multiples: juxtaposition

Juxtaposing time



Integrating temporal representations into map through glyphs

HerbariaViz,
Tom Auer et Craig McCabe, PennState University

Small multiples: juxtaposition

Juxtaposing time



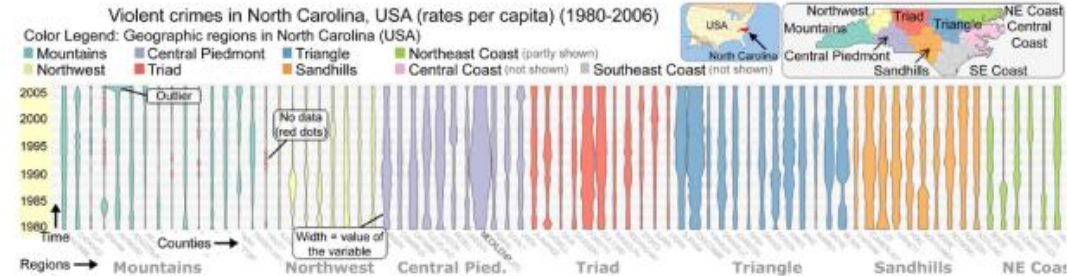
Integrating temporal
representations into map through
glyphs

+ temporal animation

Evolution du COVID-19 par département
<https://twitter.com/datagistips/status/1248508331263545344>

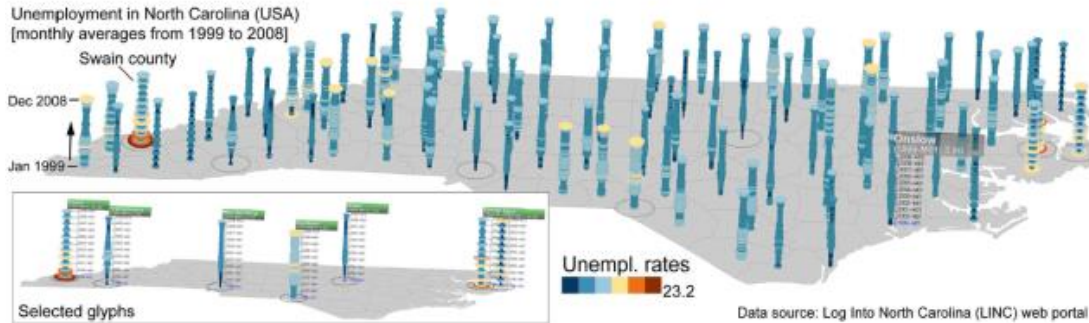
Small multiples: juxtaposition

Juxtaposing time...



...in a space-time cube

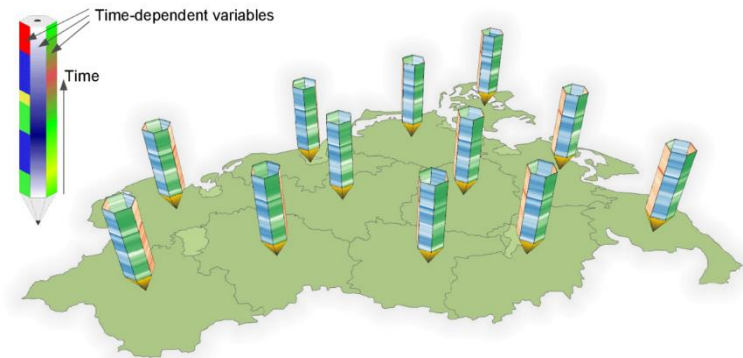
3D glyphs as temporal charts



Thakur, S. & Rhyne, T.-M, 2009

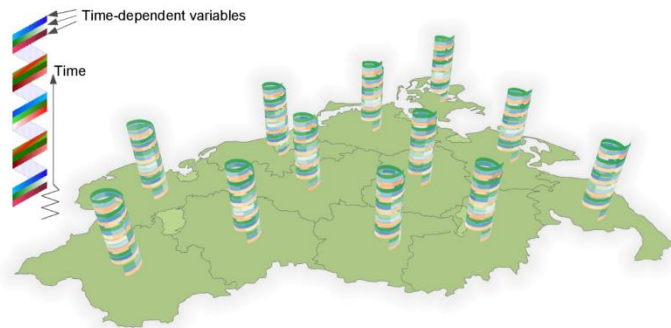
Small multiples: juxtaposition

Juxtaposing time...



...in a space-time cube

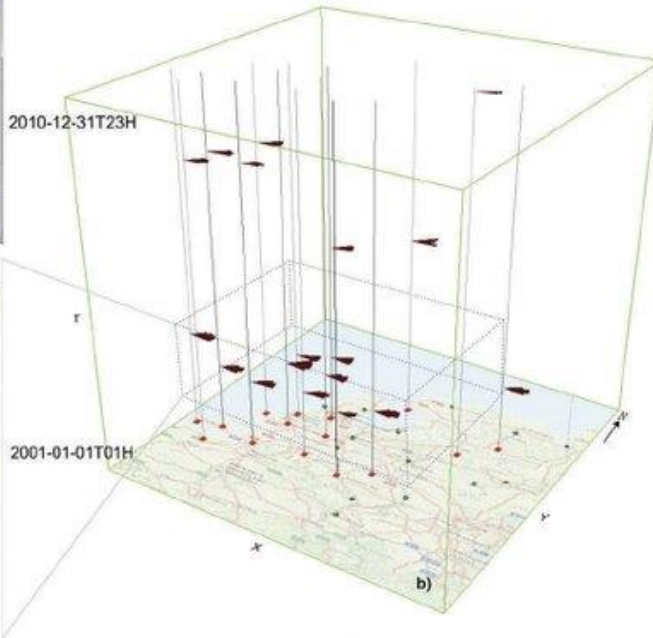
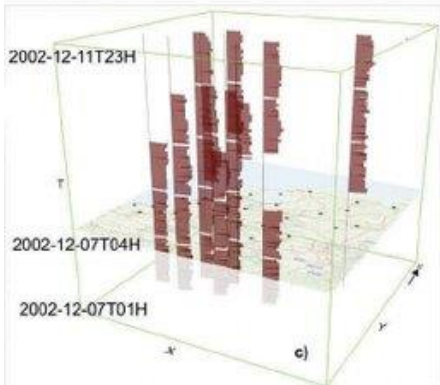
3D glyphs as temporal charts



Helix diagrams
Tominski et al., 2005

Small multiples: juxtaposition

Juxtaposing time...



...in a space-time cube

3D glyphs as temporal charts

Small multiples: juxtaposition

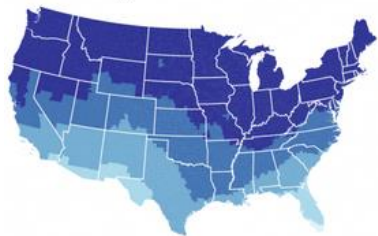
Juxtaposing space

A year of sunshine.

Avg. daily sunlight, by month, 1979-2011 (measured in kilojoules of solar radiation per square meter)



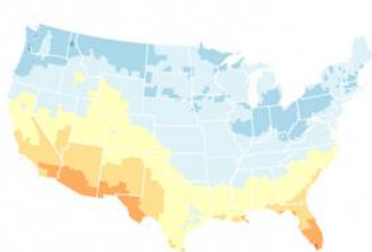
January



February



March



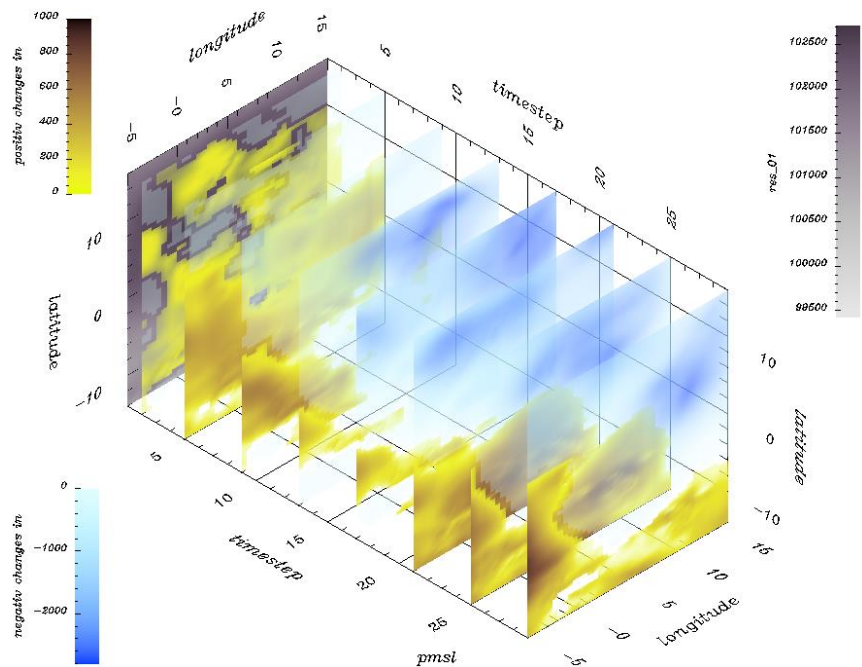
Juxtaposing space

Collection of spatial
representations

→ one per temporal primitive
(period or moment)

Small multiples: juxtaposition

Juxtaposing space...



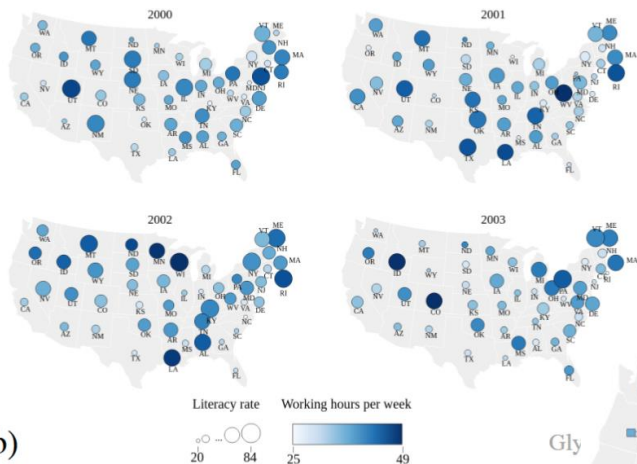
...in a space-time cube

Integrating several 2D maps for
several temporal moments

Kraak, 2003

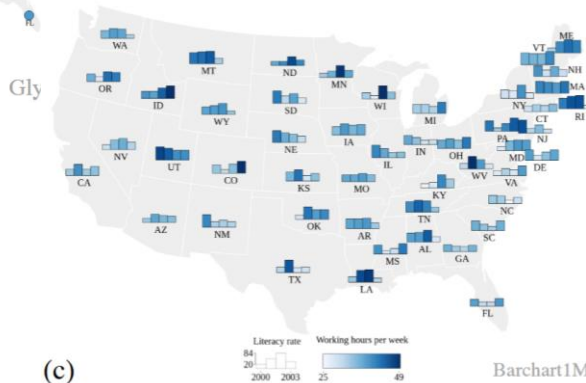
Small multiples: juxtaposition

Juxtaposing space vs juxtaposing time



Analyzing temporal evolution
→ *juxtaposing time*: glyphs

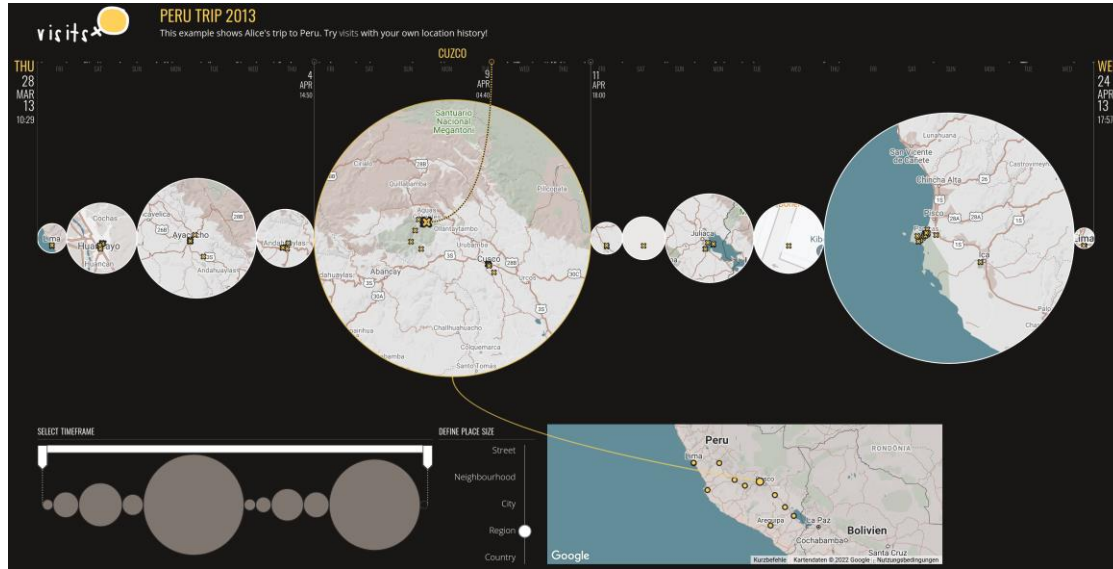
Comparing spatial distribution for
several temporal moments
→ *juxtaposing space*: collections of
maps



Peña-Araya, Vanessa, Emmanuel Pietriga et Anastasia Bezerianos. "Une comparaison des visualisations pour identifier la corrélation dans l'espace et le temps." IEEE Transactions on Visualization and Computer Graphics 26.1 (2019): 375-385.

Small multiples: juxtaposition

Juxtaposing spatio-temporal



Representing different periods and place for different chronological sequences

Offer a way to tell a story about a phenomena

Can be used with narrative patterns

Thudt, A., Baur, D., & Carpendale, S. (2013). Visits: A Spatiotemporal Visualization of Location Histories. Proceedings of the Eurographics / IEEE Conference on Visualization (EuroVis) - Short Papers.

<https://doi.org/10.2312/PE.EuroVisShort.EuroVisShort2013.079-083>

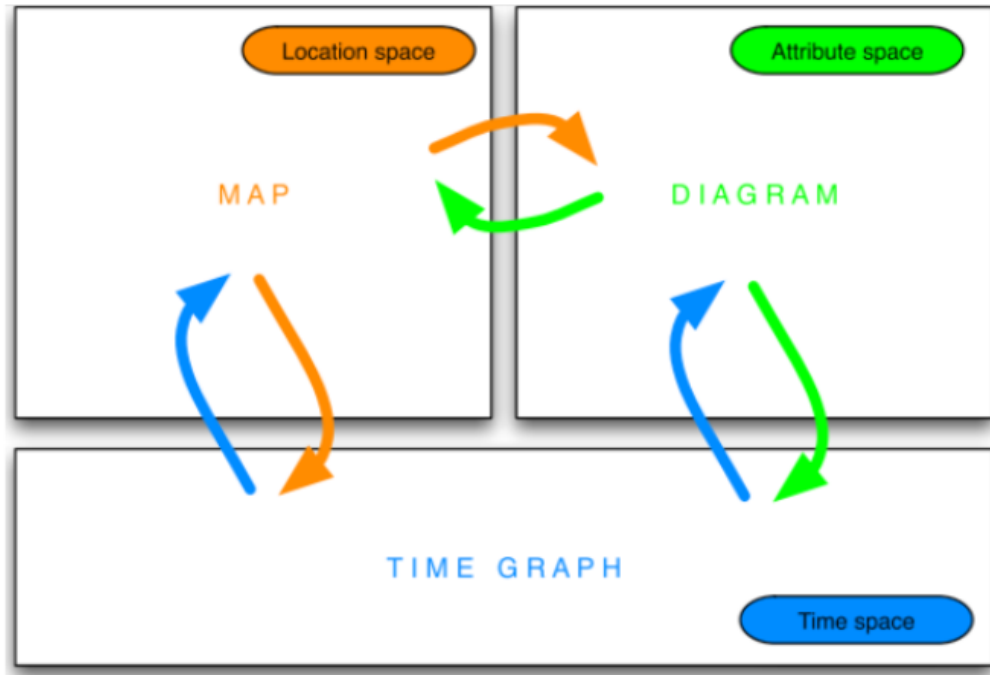
Small multiples: juxtaposition

Trend identification/comparison: comparison of spatial distribution at different moments in time; comparison of temporal evolution at several spatial locations.

Integrating charts into a map is controversial in cartography
Difficulties in reading and interpretation, especially with a large number of elements or if the entities are too far apart.

Beyond a certain number of items, not efficient

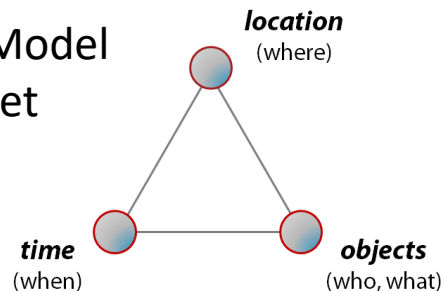
Multi-view (dashboards)



Kraak and Ormeling, 2011

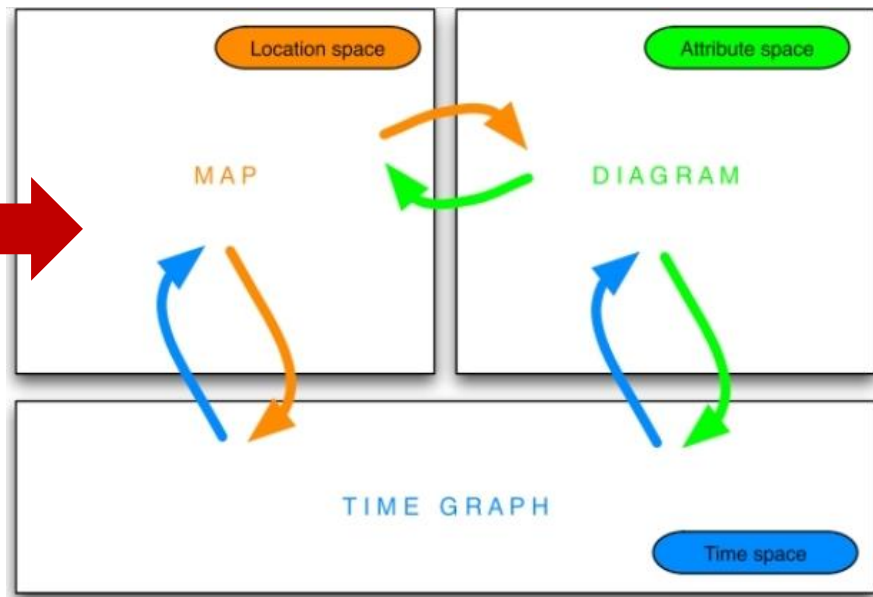
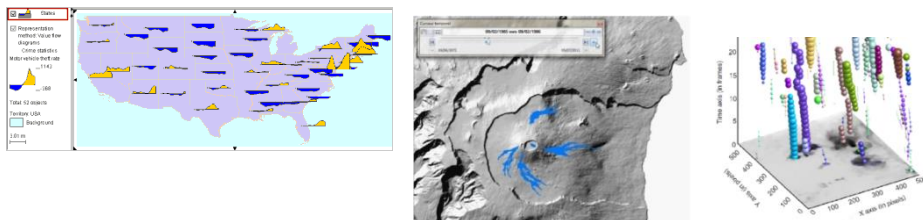
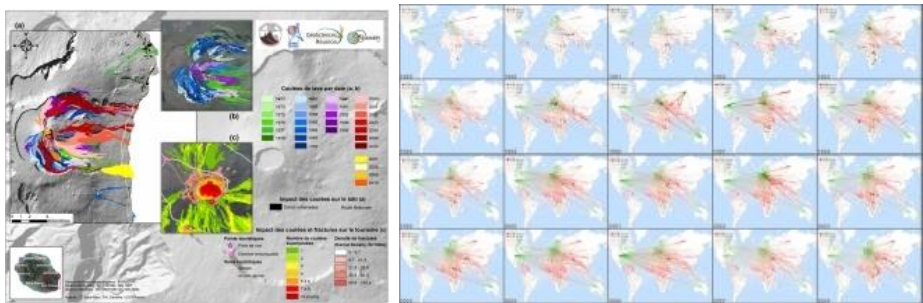
Synchronized (carto-)graphic representations

Triad Model
Peuquet
1994



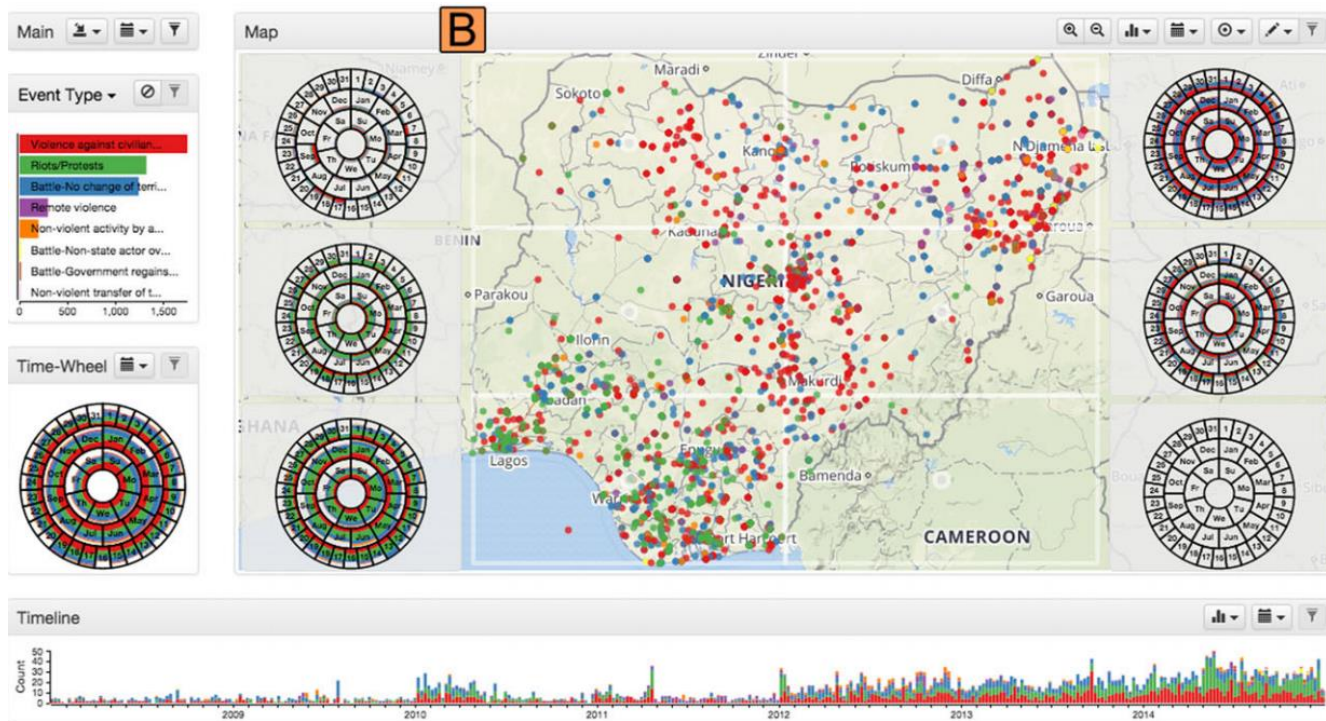
Multi-view (dashboards)

Transversal approach



Multi-view (dashboards)

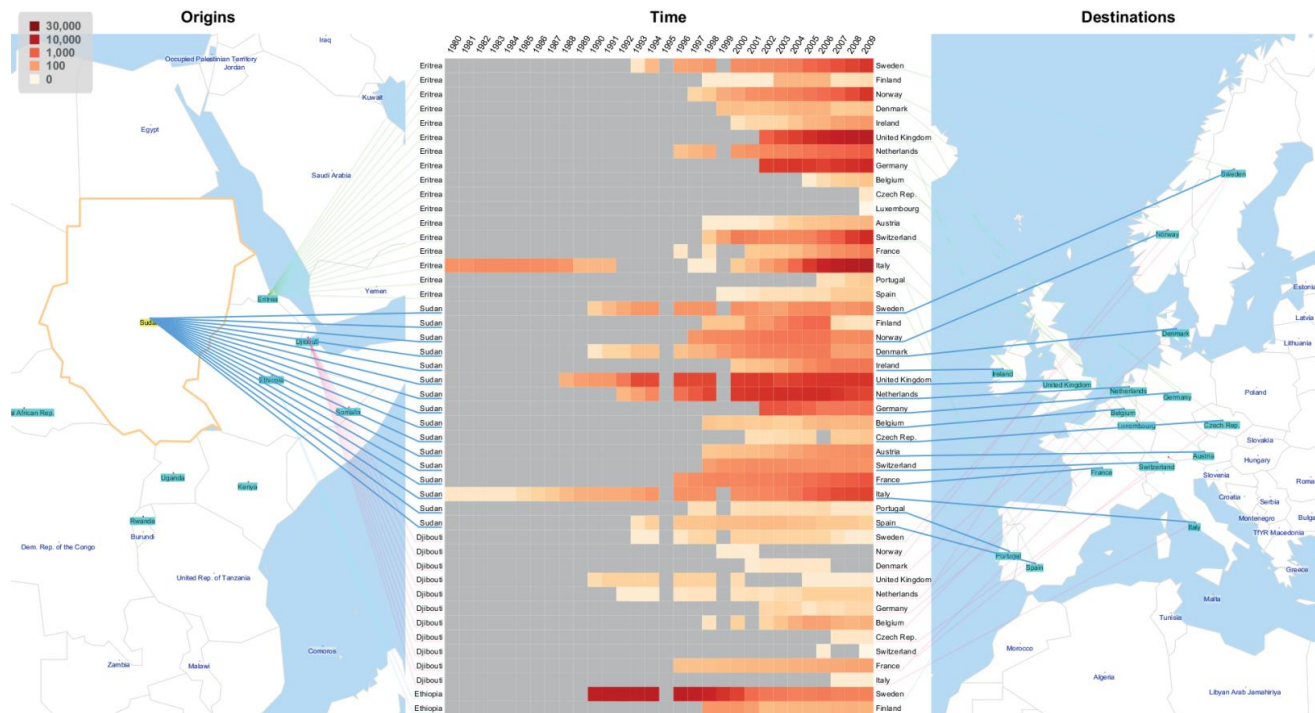
Transversal approach



Multi-vues
+
Small multiples

Perse visualization
environment
Swedberg and Peuquet,
2017

Transversal approach



Flowstrate, Boyandin 2011

Conclusion

- Important to combine temporal and spatial components in a single graphical representation to identify the spatiotemporal structure of phenomena.
- Space cannot be represented solely by a 2D map.
- Associate spatial and temporal representations
- Do not rely solely on animation.



The TimeViz Browser 2.0

A Visual Survey of Visualization Techniques for Time-Oriented Data



161/161
techniques

Filters ?

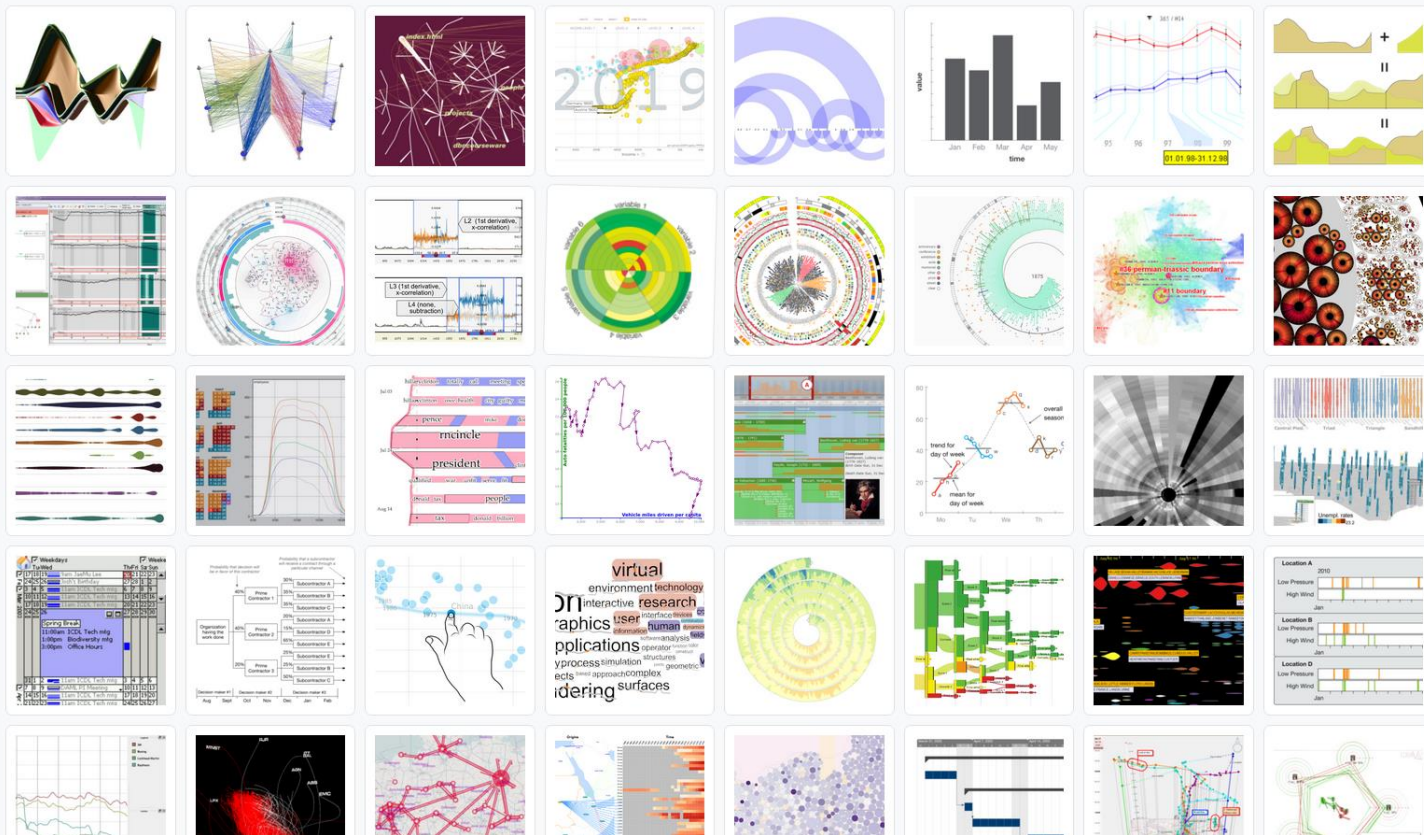
- Time
- Data
- Visual representation

Tags

- axes-based (8)
- calendar (5)
- glyph (16)
- helix (2)
- hierarchy (11)
- network (8)
- pixel-based (22)
- projections (3)
- space-time cube (9)
- sparkline (6)
- spiral (5)
- storyline (4)
- streamgraph (10)
- timeline (18)
- wordcloud (2)

Our Book

The TimeViz Browser is based on our **open-access** book on visualizing time-oriented data.



<https://browser.timeviz.net/>